

PARMA, Italy

WMO: 161300

Lat: 44.8172N

Lon: 10.2969E

Elev: 167

StdP: 14.61

Time Zone: 1.00 (EUC)

Period: 06-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.1	25.0	15.3	12.0	28.8	18.7	14.2	31.0	18.3	50.8	12.9	47.4	3.5	200	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	93.2	70.8	91.0	70.7	88.1	69.8	73.6	87.3	72.5	86.4	71.2	84.9	5.5	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.5	108.9	80.5	67.9	103.3	79.0	66.4	97.9	77.7	37.2	87.6	36.2	86.4	35.2	84.7	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.7	13.7	11.3	DB	16.7	97.6	6.2	2.1	12.3	99.2	8.7	100.4	5.2	101.6	0.7	103.1	(4)
			WB	16.4	76.1	6.5	1.1	11.7	76.9	7.9	77.6	4.3	78.2	-0.4	79.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.4	37.8	40.5	48.6	57.1	64.7	72.0	76.7	75.4	68.0	59.1	49.0	38.5
	DBStd	15.00	6.94	7.80	6.26	5.31	5.40	5.62	4.32	4.88	4.53	6.89	7.02	6.95
	HDD50	1238	387	277	100	7	0	0	0	0	0	11	94	362
	HDD65	3890	844	687	508	241	73	9	0	1	19	203	483	822
	CDD50	3931	9	11	58	220	457	661	827	788	540	292	63	5
	CDD65	1106	1	1	1	3	65	220	362	323	109	19	2	0
	CDH74	10720	0	0	9	60	583	2064	3865	3296	780	63	0	0
	CDH80	4149	0	0	0	6	132	754	1672	1403	176	6	0	0
Wind	WSAvg	4.5	3.5	4.0	5.3	5.1	5.6	5.2	5.2	4.8	4.2	3.7	3.7	3.7
Precipitation	PrecAvg	36.3	2.5	2.6	3.0	3.2	2.8	2.6	1.9	3.2	3.1	4.1	4.6	2.7
	PrecMax	46.8	5.9	6.4	10.9	8.7	5.9	7.1	5.1	7.4	7.4	12.8	8.3	5.2
	PrecMin	26.2	0.1	0.0	0.1	0.5	0.5	0.3	0.2	0.0	0.1	0.0	0.6	0.2
	PrecStd	6.1	1.8	1.6	2.4	1.9	1.7	1.8	1.4	2.3	2.0	3.5	2.0	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.4	67.4	73.2	79.1	88.2	93.0	96.0	97.0	87.8	80.2	67.5	67.5
		MCWB	49.0	55.1	54.7	61.6	68.7	71.4	70.4	70.1	69.8	63.7	61.7	64.4
	2%	DB	66.7	66.7	67.8	75.2	84.2	90.7	93.3	93.4	84.5	75.0	67.4	67.1
		MCWB	50.0	54.2	53.7	60.1	65.0	70.8	70.8	70.8	68.1	64.3	62.0	64.4
	5%	DB	49.6	54.8	64.3	72.0	80.6	87.7	91.1	90.2	81.7	71.4	67.1	53.4
		MCWB	45.0	48.0	52.4	58.5	64.1	70.3	71.0	70.2	66.1	62.8	62.5	48.2
	10%	DB	45.7	50.4	60.7	68.7	77.1	84.5	88.0	87.8	78.6	68.1	66.7	48.0
		MCWB	43.9	44.9	51.2	57.5	62.2	68.6	70.2	69.0	64.5	60.6	63.4	45.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.7	51.6	57.1	63.6	69.9	75.0	75.2	74.8	71.7	67.3	64.8	64.8
		MCDB	58.9	57.0	66.9	75.1	83.2	87.0	88.7	88.3	82.6	73.1	66.6	66.9
	2%	WB	47.2	49.5	55.5	61.4	67.3	72.9	73.7	73.3	69.8	65.4	64.7	64.4
		MCDB	49.6	54.0	64.9	71.1	79.7	86.5	87.7	87.1	80.5	70.7	66.5	66.2
	5%	WB	45.2	47.6	53.9	60.0	65.6	71.2	72.5	72.1	68.4	64.7	64.4	49.8
		MCDB	46.7	51.1	62.5	69.6	77.4	85.0	86.7	85.9	78.4	68.4	66.2	52.0
	10%	WB	43.3	46.0	52.1	58.5	63.7	69.7	71.3	70.7	66.9	64.3	64.0	45.7
		MCDB	44.7	48.7	58.6	67.8	74.6	82.6	85.3	84.2	75.7	67.5	65.8	47.3
Mean Daily Temperature Range	5% DB	MDBR	10.7	14.3	17.8	18.6	20.2	19.5	21.4	21.1	17.0	13.8	12.1	13.4
		MCDBR	18.7	19.0	24.8	25.0	24.2	23.2	24.1	24.6	22.5	18.5	17.0	24.8
		MCWBR	13.0	14.4	14.2	13.3	11.0	10.0	9.4	9.8	9.8	9.9	12.9	20.8
	5% WB	MCDBR	15.2	17.5	21.4	21.6	22.0	22.1	21.9	21.9	18.4	12.0	17.9	23.3
		MCWBR	11.1	12.9	13.0	12.0	10.9	9.9	9.1	9.5	8.7	7.3	17.5	21.5
Clear-Sky Solar Irradiance	taub	0.361	0.390	0.437	0.452	0.448	0.469	0.454	0.444	0.433	0.438	0.390	0.355	
	taud	2.334	2.220	2.094	2.099	2.156	2.145	2.194	2.236	2.238	2.169	2.255	2.367	
	Ebn at Noon	232	245	248	257	262	256	259	257	248	224	216	221	
	Edh at Noon	26	34	44	48	47	47	45	41	38	36	28	23	
All-Sky Solar Radiation	RadAvg	459	713	1116	1437	1785	2050	2114	1809	1336	802	475	393	
	RadStd	67	108	162	161	167	118	119	119	89	115	70	52	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (31 neighbors)		+0.61	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+115	+25

Nomenclature: See separate page